Electronic Devices & Circuits Laboratory
Experiment 2: BJT Characteristics

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1 Aim

- To plot the input and output characteristics of a BJT in the Common Emitter (CE) configuration, and to determine the current gain β, input resistance r_i, and output resistance r_o.
- To plot the input and output characteristics of a BJT in the Common Base (CB) configuration, and to determine the current gain α, input resistance r_i, and output resistance r_o.
- To plot the input and output characteristics of a BJT in the Common Collector (CC) configuration, and to determine the current gain γ, input resistance r_i, and output resistance r_o.
- To verify the relationships $\beta = \alpha/(1-\alpha)$ and $\gamma = \beta + 1$ from the measured data.

2 Apparatus / Components

S.No.	Component / Equipment	Specification	Quantity
1	BC547 transistor	NPN, TO-92 package	1
2	Regulated DC power supply	0-30 V, dual output	2
3	Milliammeter	0-10 mA and 0-30 mA range	2
4	Microammeter	0-500 μA range	1
5	Voltmeter	0-1 V and 0-30 V range	2
6	Fixed resistors	1 kΩ, 4.7 kΩ, 10 kΩ	as needed
7	Breadboard, connecting wires, probes	Required	as needed

Table 1: List of components required.

3 Common Emitter (CE) Configuration

3.1 Theory

A Bipolar Junction Transistor (BJT) is a three-terminal device - emitter (E), base (B), and collector (C) - formed by two back-to-back PN junctions. In normal (active-region) operation, the base-emitter junction is forward-biased and the base-collector junction is reverse-biased. A small base current controls a much larger collector current, giving current gain.

In the CE configuration, the emitter is common to both input (base-emitter) and output (collector-emitter) loops. The input characteristic (I_B vs V_{BE} at constant V_{CE}) resembles a diode's forward characteristic. The output characteristic (I_C vs V_{CE} at constant I_B) shows a steep rise in the saturation region followed by a nearly flat active region, whose slight upward slope reflects a finite output resistance. The current gain $\beta = \Delta I_C/\Delta I_B$ is typically 50-300.

3.2 Circuit Connections and Procedure

Output Characteristics:

1. Connect the transistor in CE configuration: emitter grounded (common), base current supplied through a 10 kΩ resistor from the base supply with a microammeter (0-500 μA) in series, and collector connected through a milliammeter (0-10 mA) to the collector supply.

2. Connect a voltmeter (0-1 V) across base-emitter and a voltmeter (0-30 V) across collector-emitter.

3. Set the base supply to give $I_B = 20\mu A$ (adjust and confirm on the microammeter) and keep it fixed for this run.

4. Starting with $V_{CE} = 0$ V, increase V_{CE} in steps of 0.5 V up to 2 V, then in steps of 2 V up to 20 V, recording I_C at each step.

5. Repeat for $I_B = 40, 60, 80$, and 100 μA, tabulating I_C vs V_{CE} for each. Switch off the supply.

Input Characteristics:

1. Retain the CE circuit. Set $V_{CE} = 1$ V and hold it fixed for this run.

2. Increase V_{BE} in steps of 0.05 V from 0 to 0.4 V, then in steps of 0.02 V up to 0.7 V, recording I_B at each step.

3. Repeat for $V_{CE} = 5$ V and $V_{CE} = 10$ V. Switch off the supply.

3.3 Circuit and Simulation

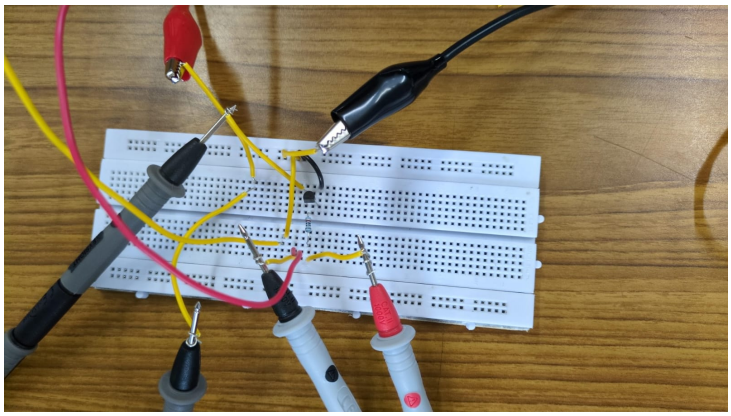


Figure 1: Hardware setup for CE configuration

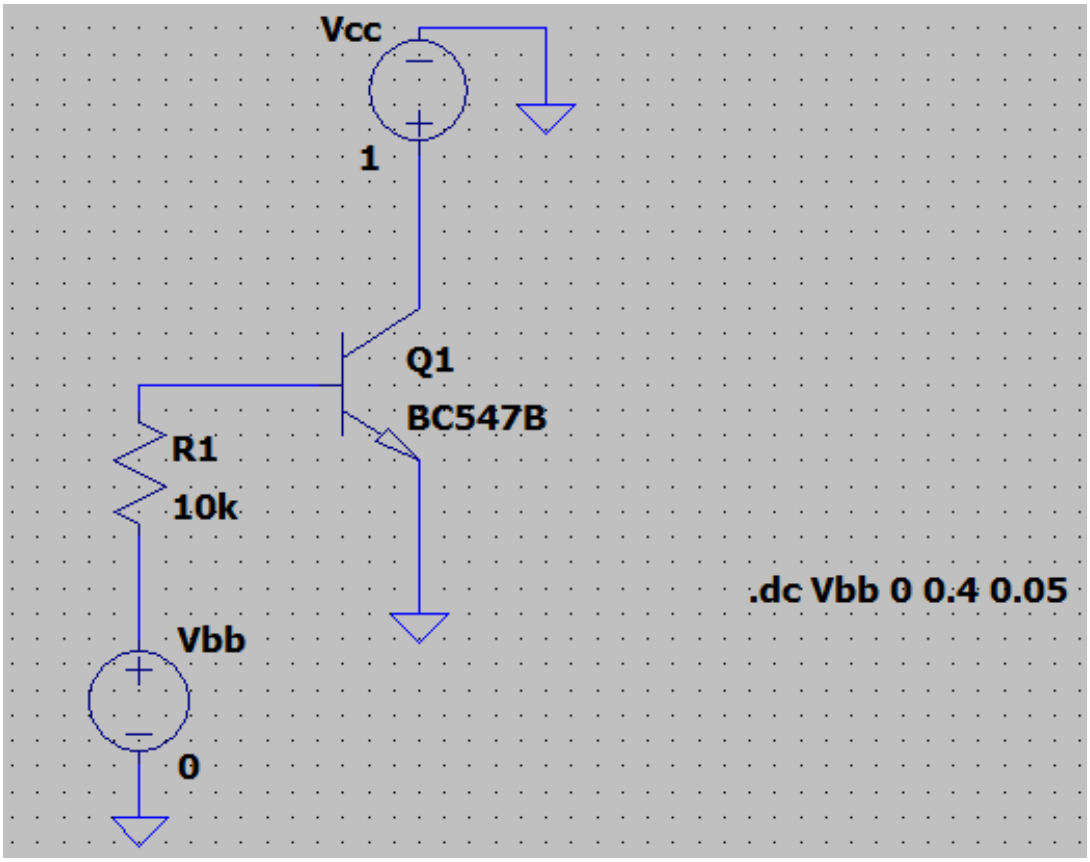


Figure 2: Simulation Circuit for CE Input Characteristics

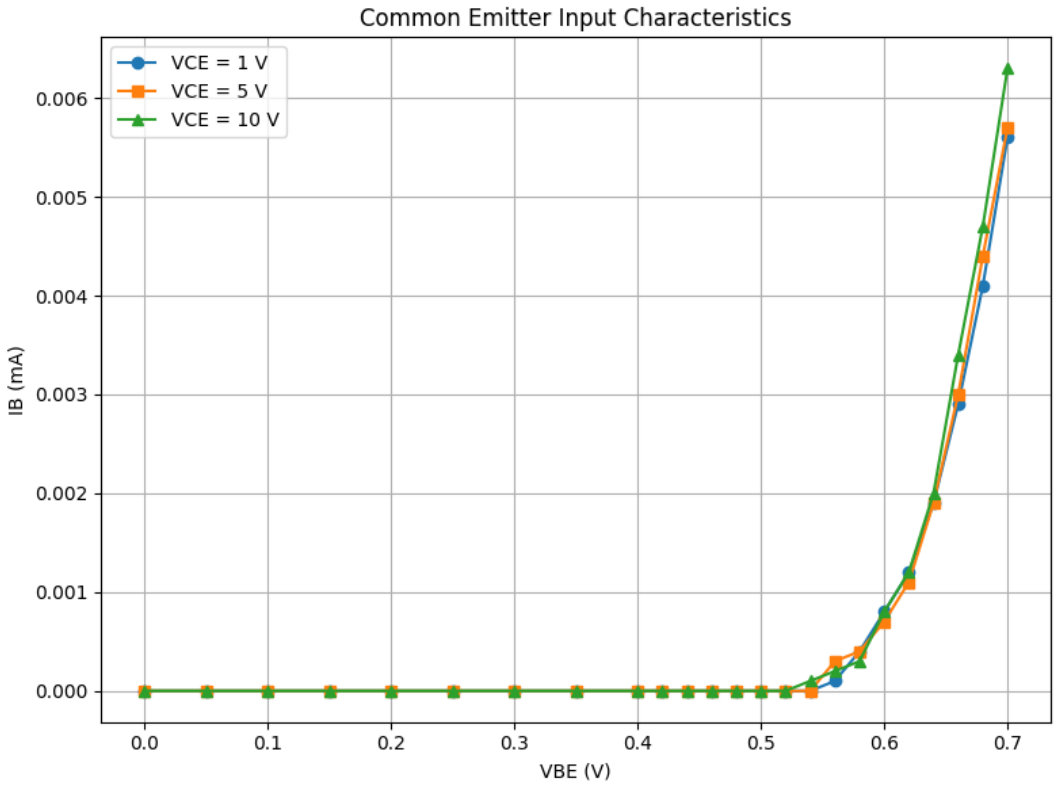


Figure 3: Simulated CE Input Characteristics (I_B vs V_{BE})

3.3.1 CE Input Simulations

The following simulations show the input characteristics at varying V_{CE} values.

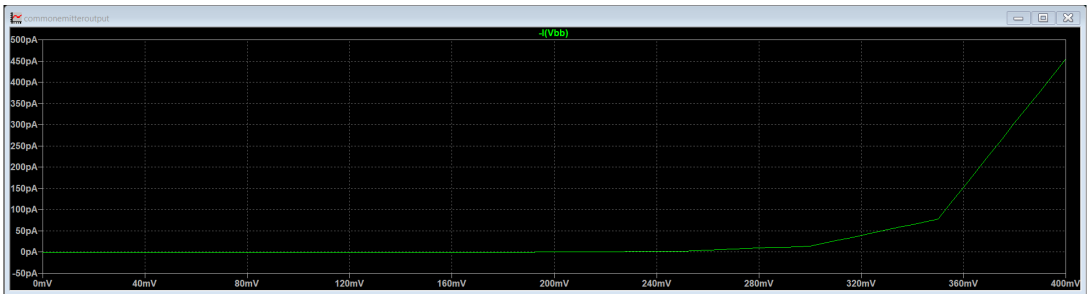


Figure 4: CE Input $V_{CE} = 1V$ (Low Range)

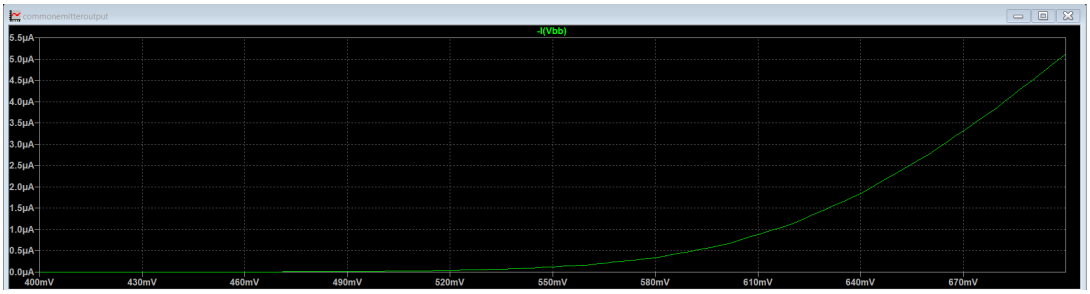


Figure 5: CE Input $V_{CE} = 1V$ (High Range)

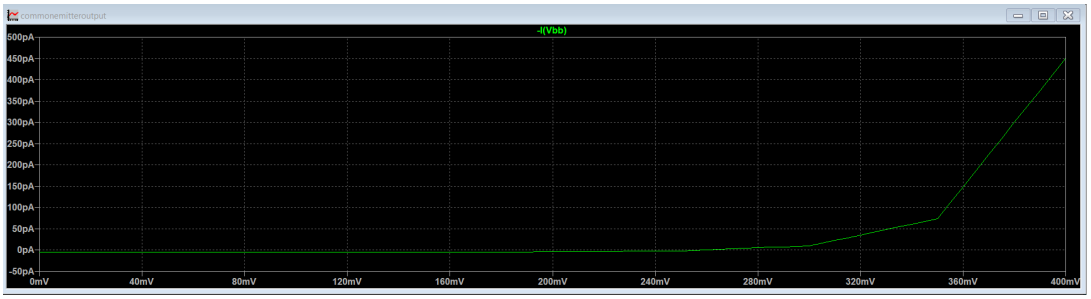


Figure 6: CE Input $V_{CE} = 5V$ (Low Range)

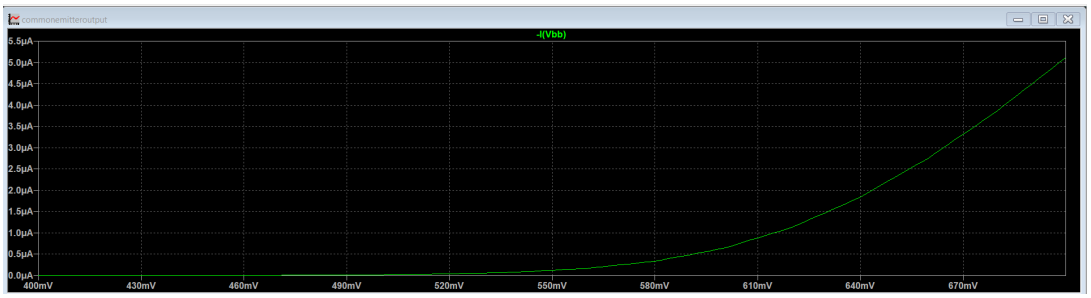


Figure 7: CE Input $V_{CE} = 5V$ (High Range)

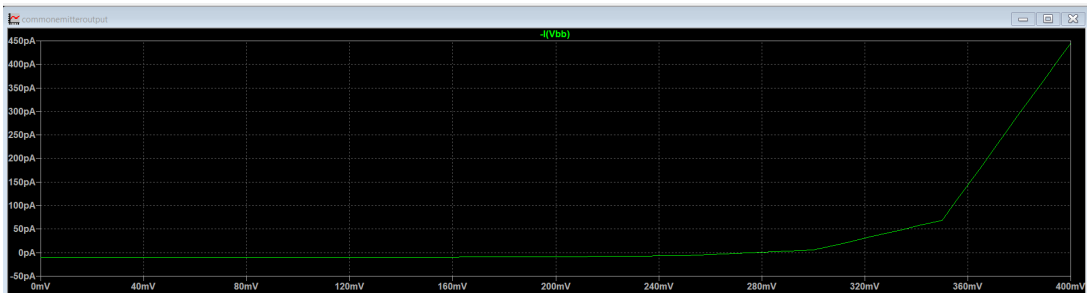


Figure 8: CE Input $V_{CE} = 10V$ (Low Range)

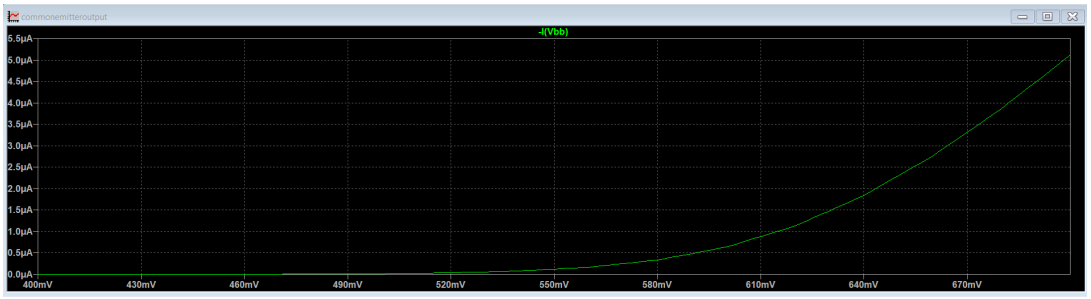


Figure 9: CE Input $V_{CE} = 10V$ (High Range)

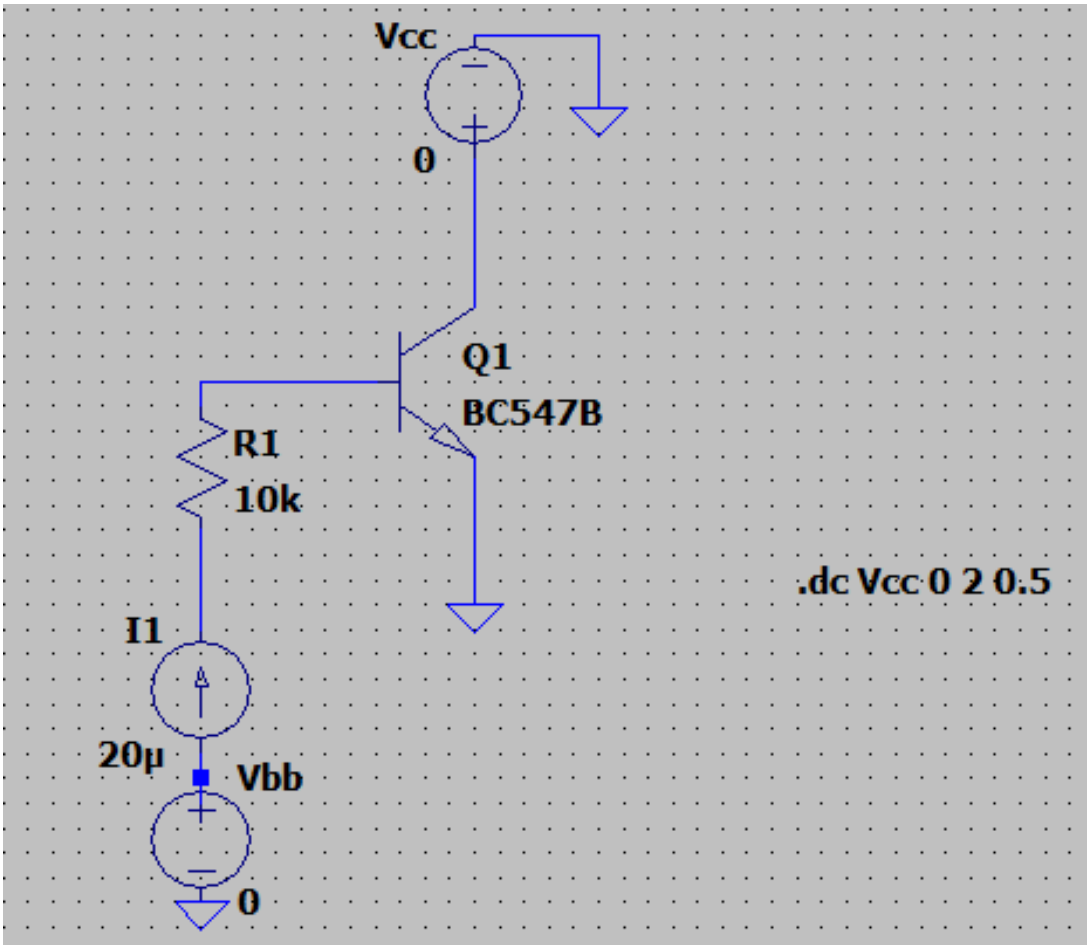


Figure 10: Simulation Circuit for CE Output Characteristics

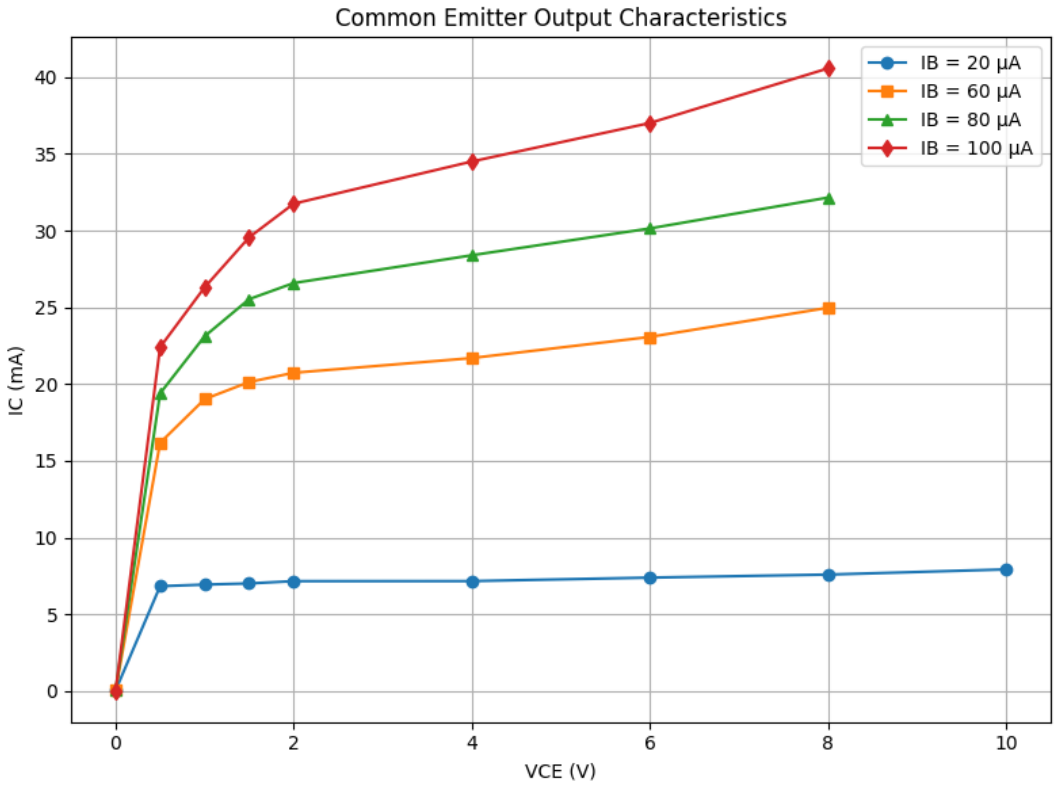


Figure 11: Simulated CE Output Characteristics (I_C vs V_{CE})

There is a slight increase in the active region in the I_c value due to the increase in V_{CE} which causes an reverse bias increase in the BC junction, which widens the depletion layer making the base thinner.

3.3.2 CE Output Simulations

The following images show the simulation results for varying base currents. The 'a' images represent the voltage sweep in steps of 0.5 V up to 2 V, and 'b' images represent steps of 2 V up to 20 V.

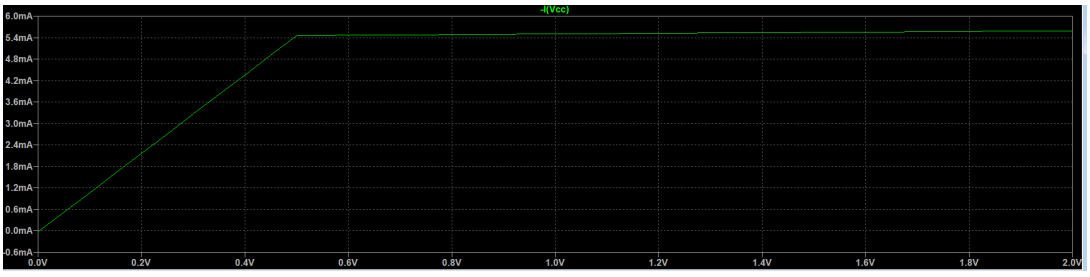


Figure 12: CE Output $I_B = 20\mu A$ (0 to 2V)

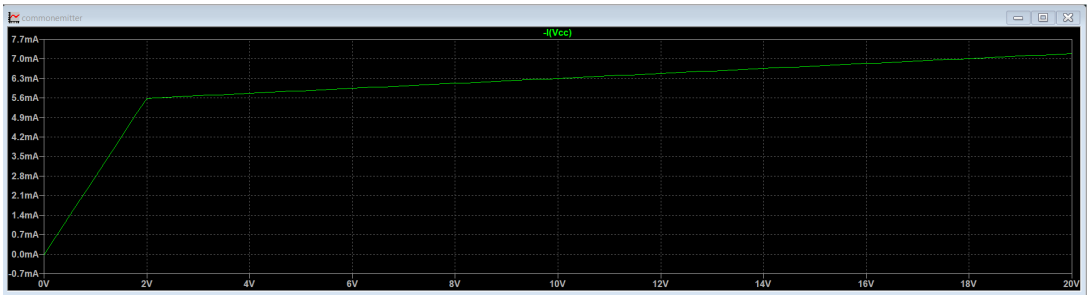


Figure 13: CE Output $I_B = 20\mu A$ (2 to 20V)

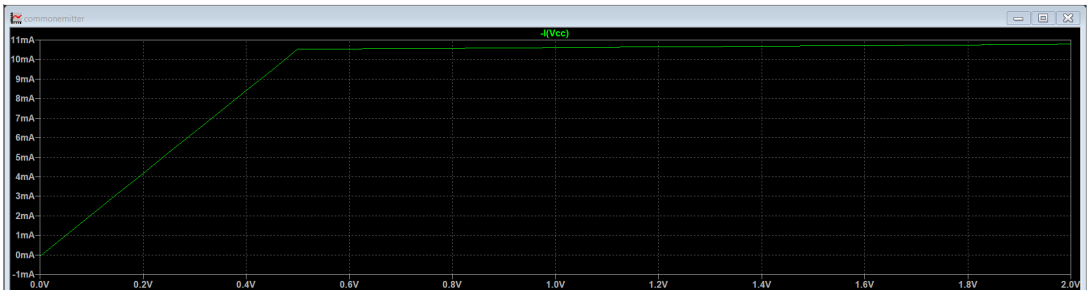


Figure 14: CE Output $I_B = 40\mu A$ (0 to 2V)

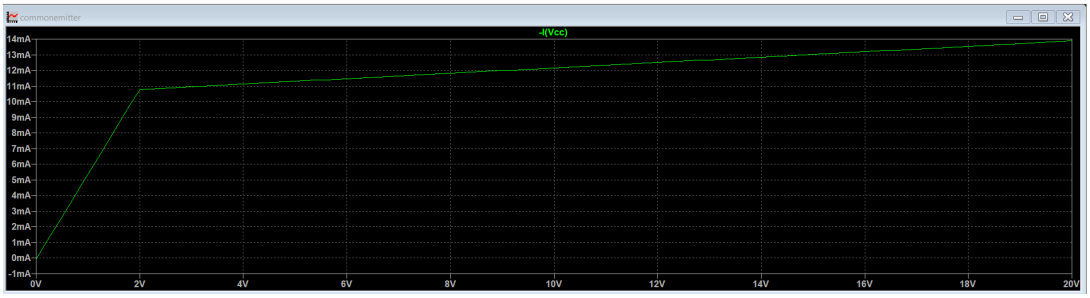


Figure 15: CE Output $I_B = 40\mu A$ (2 to 20V)

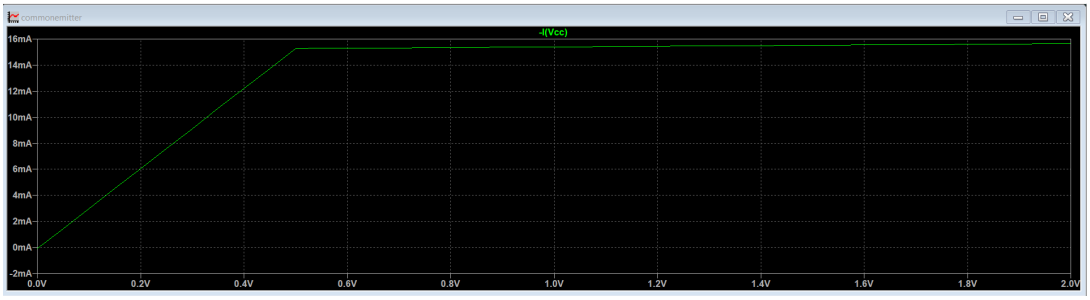


Figure 16: CE Output $I_B = 60\mu A$ (0 to 2V)

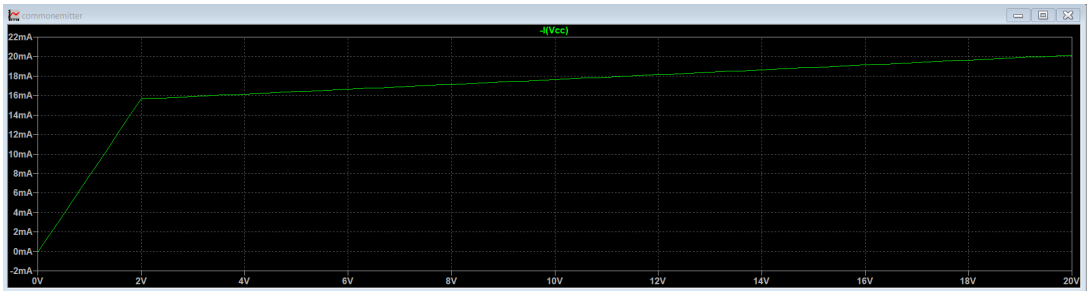


Figure 17: CE Output $I_B = 60\mu A$ (2 to 20V)

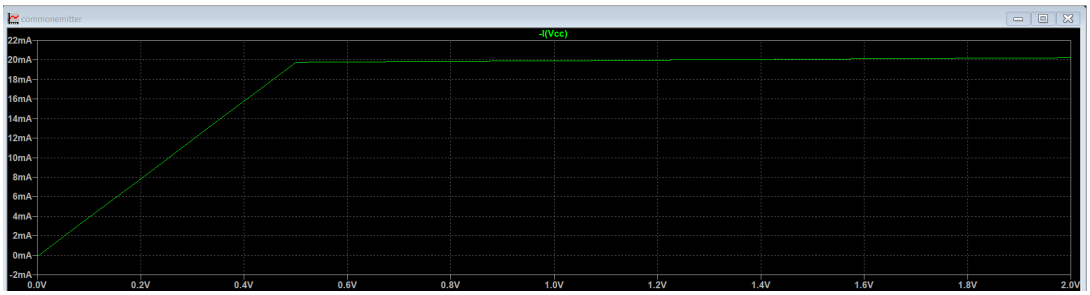


Figure 18: CE Output $I_B = 80\mu A$ (0 to 2V)

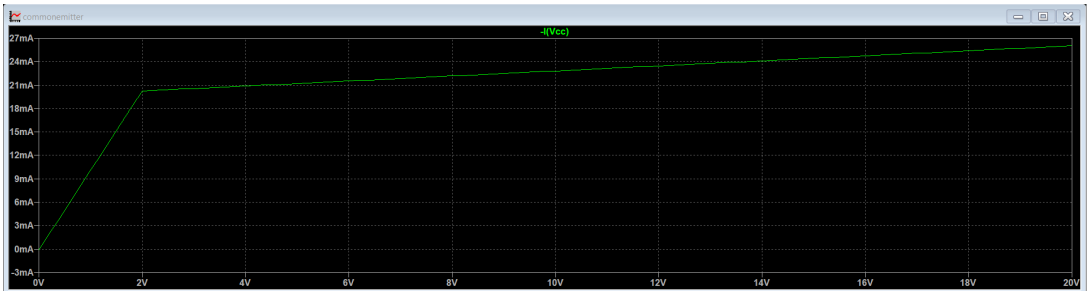


Figure 19: CE Output $I_B = 80\mu A$ (2 to 20V)

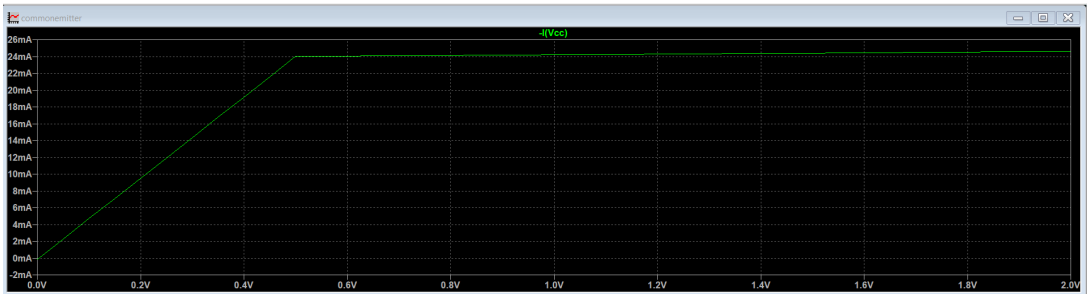


Figure 20: CE Output $I_B = 100\mu A$ (0 to 2V)

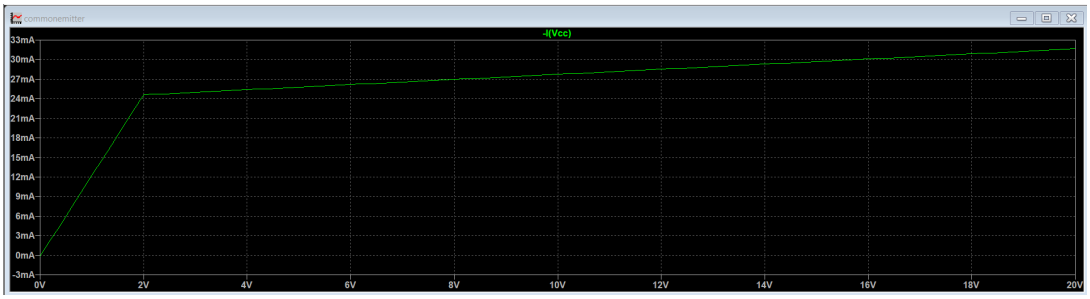


Figure 21: CE Output $I_B = 100\mu A$ (2 to 20V)

3.4 Observation Tables

Table 2: CE Output Characteristics

S.No.	V_{CE} **(V)**	I_C **(mA) at** $I_B = 20\mu A$	I_C **(mA) at** $I_B = 60\mu A$	I_C **(mA) at** $I_B = 100\mu A$
1	0.0	0.031	0.031	0.0026
2	0.5	6.825	16.170	22.4000
3	1.0	6.930	19.015	26.3010
4	1.5	7.005	20.132	29.5600
5	2.0	7.155	20.732	31.7500
6	4.0	7.158	21.691	34.5000
7	6.0	7.385	23.070	37.0100
8	8.0	7.583	24.970	40.5700

Table 3: CE Input Characteristics

S.No.	V_{BE} **(V)**	I_B **(mA) at** $V_{CE} = 1V$	I_B **(mA) at** $V_{CE} = 5V$	I_B **(mA) at** $V_{CE} = 10V$
1	0.00	0	0	0
2	0.05	0	0	0
3	0.10	0	0	0
4	0.15	0	0	0
5	0.20	0	0	0
6	0.25	0	0	0
7	0.30	0	0	0
8	0.35	0	0	0
9	0.40	0	0	0
10	0.42	0	0	0
11	0.44	0	0	0
12	0.46	0	0	0
13	0.48	0	0	0
14	0.50	0	0	0
15	0.52	0	0	0
16	0.54	0	0	0.0001
17	0.56	0.0001	0.0003	0.0002
18	0.58	0.0004	0.0004	0.0003
19	0.60	0.0008	0.0007	0.0008
20	0.62	0.0012	0.0011	0.0012
21	0.64	0.0019	0.0019	0.0020
22	0.66	0.0029	0.0030	0.0034
23	0.68	0.0041	0.0044	0.0047
24	0.70	0.0056	0.0057	0.0063

3.5 Calculations from Observation Data

Current gain β: Using the CE output table at $V_{CE} = 4.0$ V (active region, away from the saturation knee):

$$\beta_{20\to60\mu A} = \frac{\Delta I_C}{\Delta I_B} = \frac{21.691 - 7.158 \text{ mA}}{40\ \mu A} \approx 363$$

$$\beta_{60\to100\mu A} = \frac{34.500 - 21.691 \text{ mA}}{40\ \mu A} \approx 320$$

$$\beta_{20\to100\mu A} = \frac{34.500 - 7.158 \text{ mA}}{80\ \mu A} \approx 342$$

The measured β is thus in the range **320–363**, consistent with the BC547 datasheet range (110–800 depending on the sample, typically 200–400).

Output resistance r_o: From the same table, taking ΔV_{CE} between 4.0 V and 8.0 V at fixed I_B:

- At $I_B = 20\ \mu A$: $r_o = \dfrac{8.0 - 4.0 \text{ V}}{(7.583 - 7.158) \text{ mA}} \approx 9.4 \text{ k}\Omega$
- At $I_B = 60\ \mu A$: $r_o = \dfrac{4.0 \text{ V}}{(24.970 - 21.691) \text{ mA}} \approx 1.22 \text{ k}\Omega$

- At $I_B = 100\ \mu A$: $r_o = \dfrac{4.0\text{ V}}{(40.570 - 34.500)\text{ mA}} \approx 0.66\text{ k}\Omega$

r_o decreases as I_B (and hence I_C) increases, as expected from the Early-effect slope of the output family.

Input resistance r_i: Using the CE input table, in the exponential (forward-conducting) region between $V_{BE} = 0.64$ V and 0.70 V:

$$r_{i,\,V_{CE}=1V} = \frac{\Delta V_{BE}}{\Delta I_B} = \frac{0.70 - 0.68\text{ V}}{(0.0056 - 0.0041)\text{ mA}} \approx 13.3\text{ k}\Omega$$

$$r_{i,\,V_{CE}=5V} = \frac{0.02\text{ V}}{(0.0057 - 0.0044)\text{ mA}} \approx 15.4\text{ k}\Omega$$

$$r_{i,\,V_{CE}=10V} = \frac{0.02\text{ V}}{(0.0063 - 0.0047)\text{ mA}} \approx 12.5\text{ k}\Omega$$

Taking the step before that ($V_{BE} = 0.64 \rightarrow 0.66$ V) gives r_i in the range 14–20 kΩ across the three curves, so overall $r_i \approx$ 13–20 kΩ near cut-in, decreasing towards the lower end of that range as V_{BE} increases further (consistent with the diode-like I_B–V_{BE} curve becoming steeper, i.e. lower dynamic resistance, at higher forward bias). This is a moderate input resistance, as expected for CE — higher than CB ($\approx 830\ \Omega$,) but much lower than CC (≈ 8.3 MΩ), matching the expected ordering $r_{i,CB} < r_{i,CE} < r_{i,CC}$.

4 Common Base (CB) Configuration

4.1 Theory

In the CB configuration, the base is common to both loops. The input characteristic (I_E vs V_{EB} at constant V_{CB}) is again diode-like. The output characteristic (I_C vs V_{CB} at constant I_E) is extremely flat, since the collector current is set almost entirely by the emitter current and is nearly independent of V_{CB} giving CB circuits a very high output resistance. The current gain $\alpha = \Delta I_C / \Delta I_E$ is always slightly less than 1 (typically 0.95-0.99).

4.2 Circuit Connections and Procedure

Output Characteristics:

1. Connect the transistor in CB configuration: base grounded (common), emitter current supplied through a 1 kΩ resistor from the emitter supply with a milliammeter (0-10 mA) in series, and collector connected through a milliammeter (0-10 mA) to the collector supply.

2. Connect a voltmeter (0-1 V) across emitter-base and a voltmeter (0-30 V) across collector-base.

3. Set the emitter supply to give $I_E = 2$ mA and hold it fixed for this run.

4. Starting with $V_{CB} = 0$ V, increase V_{CB} in steps of 0.5 V up to 2 V, then in steps of 2 V up to 20 V, recording I_C at each step.

5. Repeat for $I_E = 4, 6, 8$, and 10 mA. Switch off the supply.

Input Characteristics:

1. Retain the CB circuit. Set $V_{CB} = 0$ V and hold it fixed for this run.
2. Increase V_{EB} in steps of 0.05 V from 0 to 0.4 V, then in steps of 0.02 V up to 0.7 V, recording I_E at each step.
3. Repeat for $V_{CB} = 5$ V and $V_{CB} = 10$ V. Switch off the supply.

4.3 Circuit and Simulation

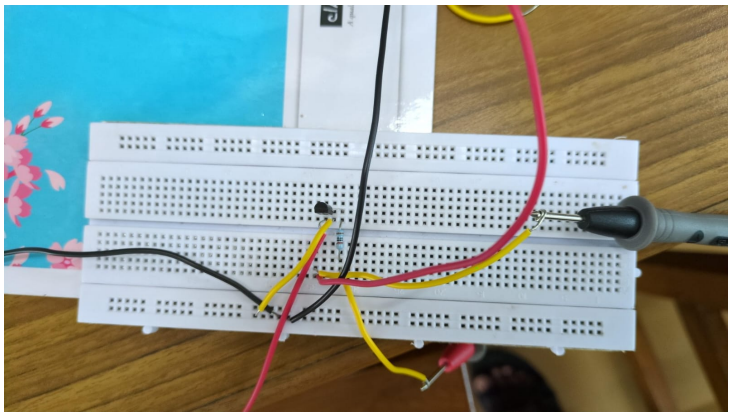


Figure 22: Hardware setup for CB configuration

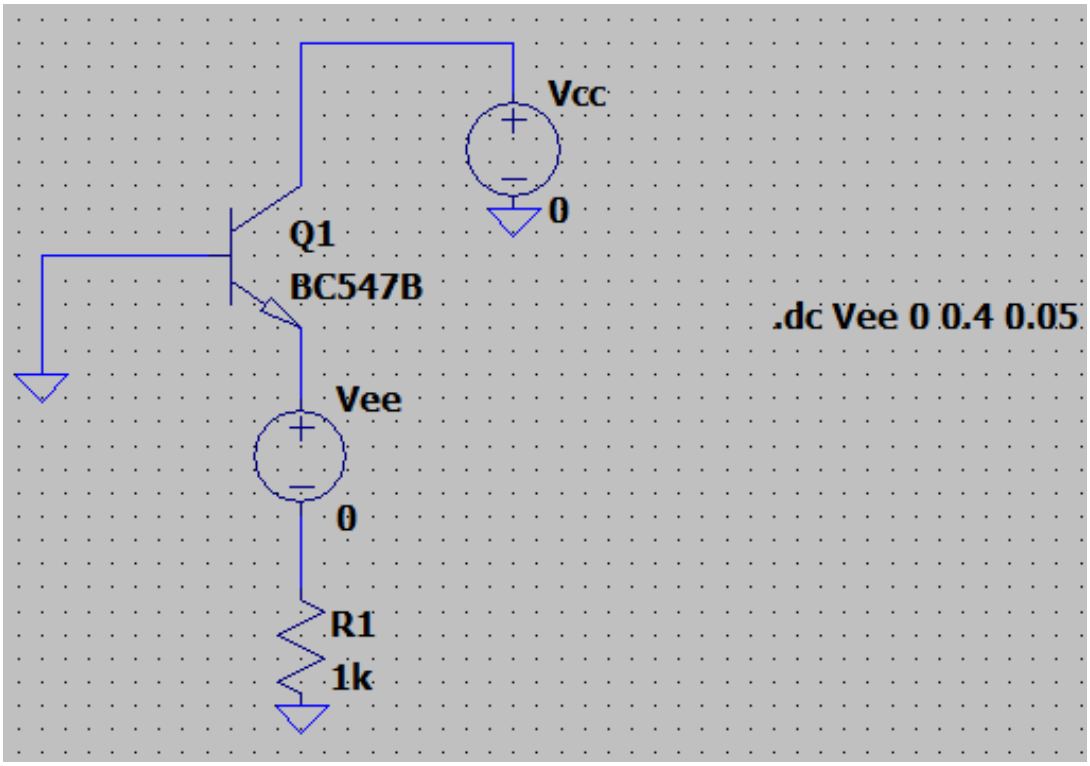


Figure 23: Simulation Circuit for CB Configuration

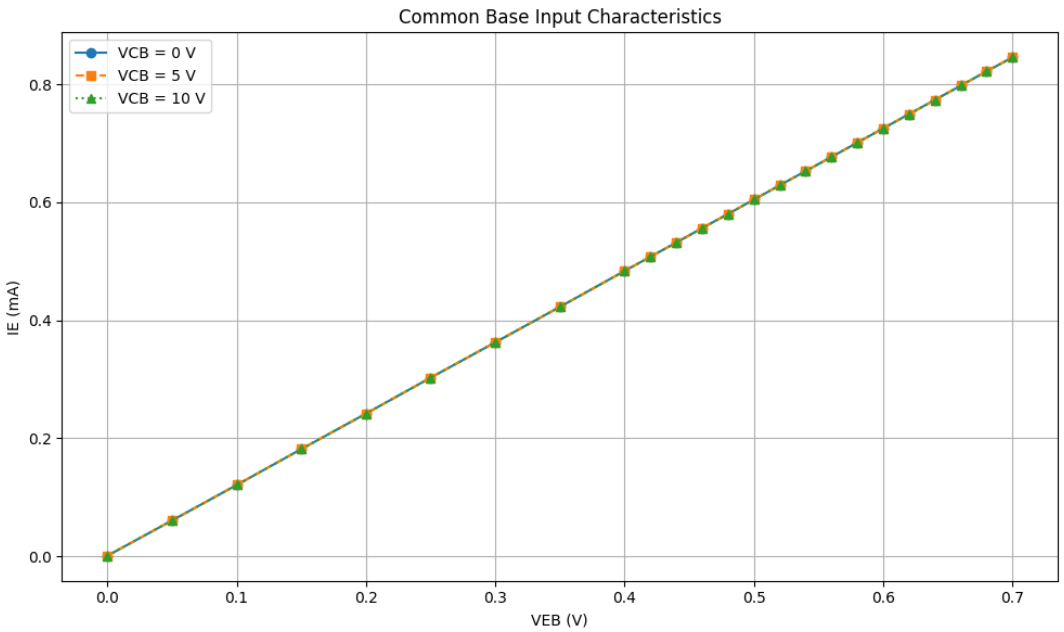


Figure 24: Simulated CB Input Characteristics (I_E vs V_{EB})

4.3.1 CB Input Simulations

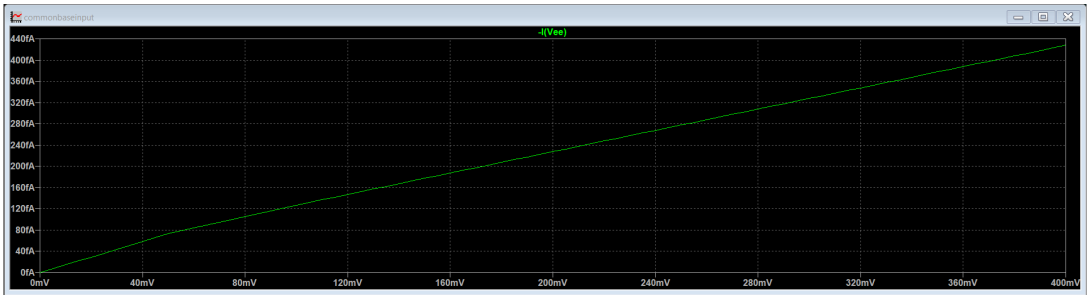


Figure 25: CB Input $V_{CB} = 0V$ (Low Range)

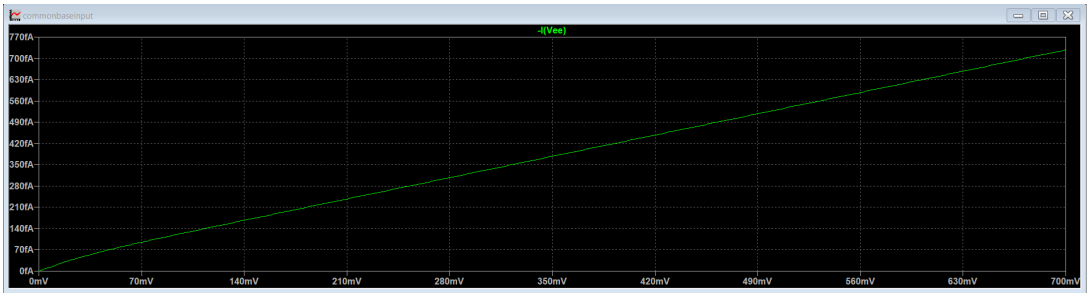


Figure 26: CB Input $V_{CB} = 0V$ (High Range)

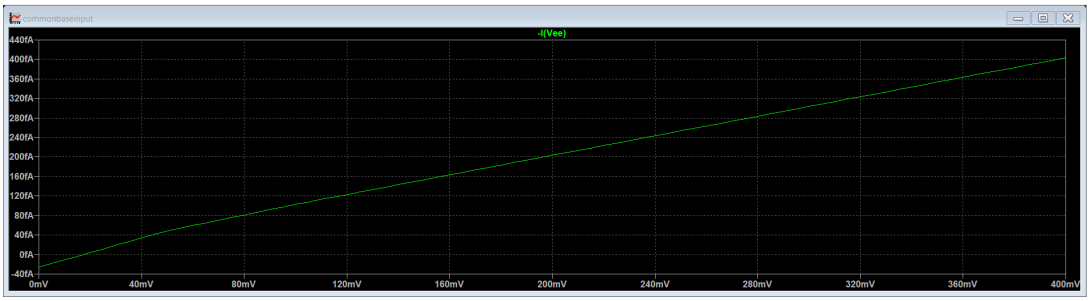


Figure 27: CB Input $V_{CB} = 5V$ (Low Range)

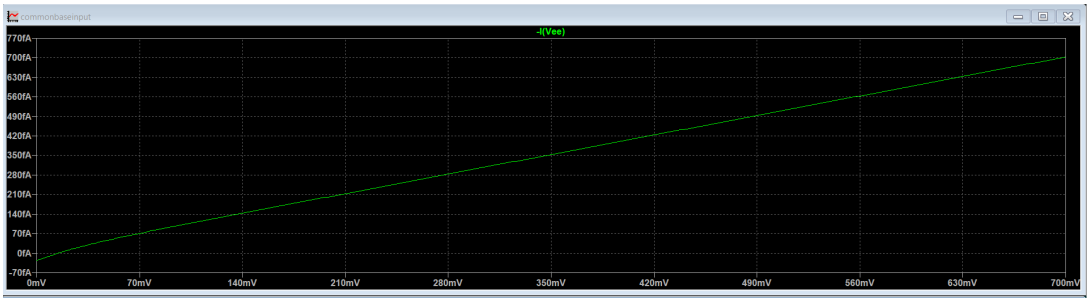


Figure 28: CB Input $V_{CB} = 5V$ (High Range)

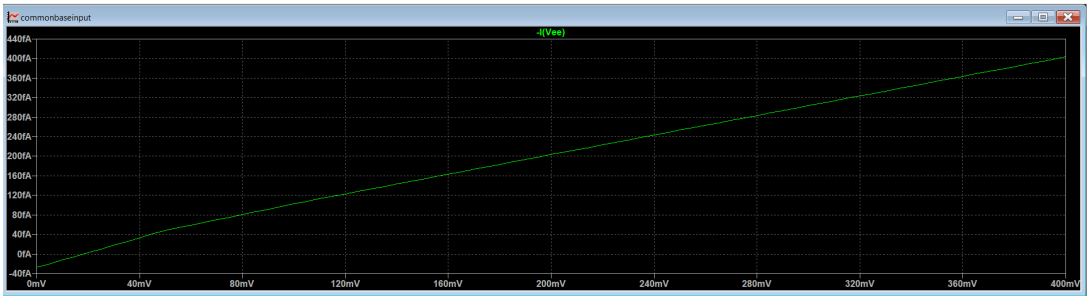


Figure 29: CB Input $V_{CB} = 10V$ (Low Range)

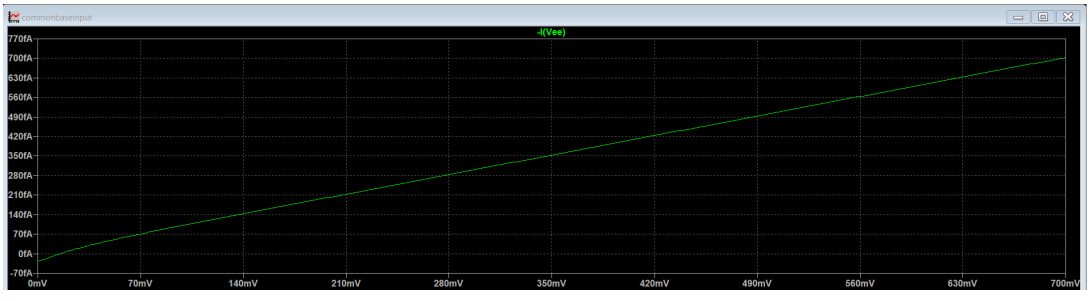


Figure 30: CB Input $V_{CB} = 10V$ (High Range)

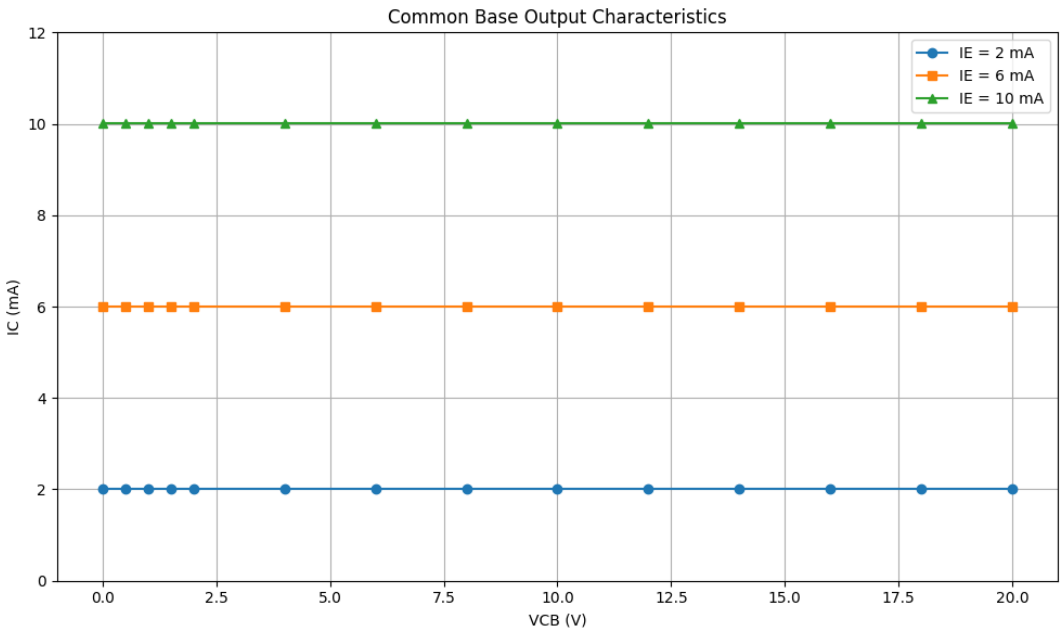


Figure 31: Simulated CB Output Characteristics (I_C vs V_{CB})

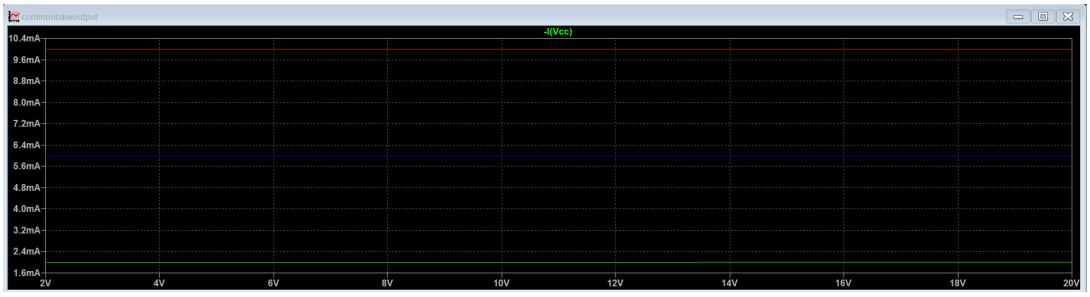


Figure 32: Simulated CB Output Characteristics Detailed Simulation

4.4 Observation Tables

Table 4: CB Output Characteristics

S.No.	V_{CB} **(V)**	I_C **(mA) at** $I_E = 2$ **mA**	I_C **(mA) at** $I_E = 6$ **mA**	I_C **(mA) at** $I_E = 10$ **mA**
1	0.0	1.99	5.98	9.96
2	0.5	1.99	5.98	9.96
3	1.0	1.99	5.98	9.96
4	1.5	1.99	5.98	9.97
5	2.0	2.00	5.98	9.97
6	4.0	2.00	5.98	9.98
7	6.0	2.00	5.99	9.98
8	8.0	2.00	5.99	9.98

Table 5: CB Input Characteristics

S.No.	V_{EB} **(V)**	I_E **(mA) at** $V_{CB} = 0V$	I_E **(mA) at** $V_{CB} = 5V$	I_E **(mA) at** $V_{CB} = 10V$
1	0.00	0.0007	0.0008	0.0008
2	0.05	0.0606	0.0606	0.0606
3	0.10	0.1207	0.1207	0.1207
4	0.15	0.1816	0.1816	0.1816
5	0.20	0.2417	0.2417	0.2417
6	0.25	0.3026	0.3026	0.3026
7	0.30	0.3625	0.3625	0.3625
8	0.35	0.4227	0.4227	0.4227

4.5 Calculations from Observation Data

Current gain α: Using the CB output table,

$$\alpha = \frac{\Delta I_C}{\Delta I_E} = \frac{9.96 - 1.99 \text{ mA}}{10 - 2 \text{ mA}} = 0.99625$$

$$\beta = \frac{0.99625}{1 - 0.99625} \approx 266$$

Output resistance r_o: Using the re-measured CB output table, taking the full swept range $V_{CB} = 0$ V to 8.0 V at each fixed I_E:

$$r_{o,\, I_E = 2mA} = \frac{\Delta V_{CB}}{\Delta I_C} = \frac{8.0 \text{ V}}{(2.00 - 1.99) \text{ mA}} \approx 800 \text{ k}\Omega$$

$$r_{o,\, I_E = 6mA} = \frac{8.0 \text{ V}}{(5.99 - 5.98) \text{ mA}} \approx 800 \text{ k}\Omega$$

$$r_{o,\, I_E = 10mA} = \frac{8.0 \text{ V}}{(9.98 - 9.96) \text{ mA}} \approx 400 \text{ k}\Omega$$

So r_o is in the range **400–800 k**Ω, roughly two orders of magnitude higher than the CE-measured r_o (0.66–9.4 kΩ) and even higher than the CC-measured r_o (2.8–7.5 kΩ). This is consistent with CB theory: since I_C is set almost entirely by I_E and depends only very weakly on V_{CB}, the CB output characteristic is the flattest of the three configurations, giving by far the highest output resistance.

Input resistance r_i: Using the CB input table, in the forward-conducting region between $V_{EB} = 0.30$ V and 0.35 V:

$$r_i = \frac{\Delta V_{EB}}{\Delta I_E} = \frac{0.35 - 0.30 \text{ V}}{(0.4227 - 0.3625) \text{ mA}} \approx 831 \ \Omega$$

Cross-check of $\beta = \alpha/(1 - \alpha)$: The CE-measured β (320–363) and the CB-derived β from $\alpha = 0.99625$ (≈ 266) agree to within the same order of magnitude, which is a reasonable cross-check given how sensitive $\beta = \alpha/(1 - \alpha)$ is to small errors in α near 1.

5 Common Collector (CC) Configuration

5.1 Theory

In the CC configuration (also called an emitter follower), the collector is common to both loops. The output characteristic (I_E vs V_{EC} at constant I_B) closely resembles the CE output characteristic, since $I_E \approx I_C$. The input characteristic is normally plotted as I_B vs V_{BC} at constant V_{EC}, and falls off as the base-collector reverse bias increases. The current gain $\gamma = \Delta I_E / \Delta I_B \approx \beta + 1$, so CC circuits provide current gain close to that of the CE configuration but are used mainly for their high input resistance and low output resistance (voltage buffering).

5.2 Circuit Connections and Procedure

Output Characteristics:

1. Connect the transistor in CC configuration: collector grounded (common), base current supplied through a 10 kΩ resistor from the base supply with a microammeter (0-500 μA) in series, and emitter connected through a milliammeter (0-10 mA) to the emitter supply.

2. Connect a voltmeter (0-30 V) across base-collector and a voltmeter (0-30 V) across emitter-collector.

3. Set the base supply to give $I_B = 20\mu A$ and hold it fixed for this run.

4. Starting with $V_{EC} = 0$ V increase V_{EC} in steps of 0.5 V up to 2 V, then in steps of 2 V up to 20 V, recording I_E at each step.

5. Repeat for $I_B = 40, 60, 80$, and 100 μA. Switch off the supply.

Input Characteristics:

1. Retain the CC circuit. Set $V_{EC} = 1$ V and hold it fixed for this run.

2. Increase V_{BC} in steps of 0.5 V from 0 to 10 V, recording I_B at each step.

3. Repeat for $V_{EC} = 5$ V and $V_{EC} = 10$ V. Switch off the supply.

5.3 Circuit and Simulation

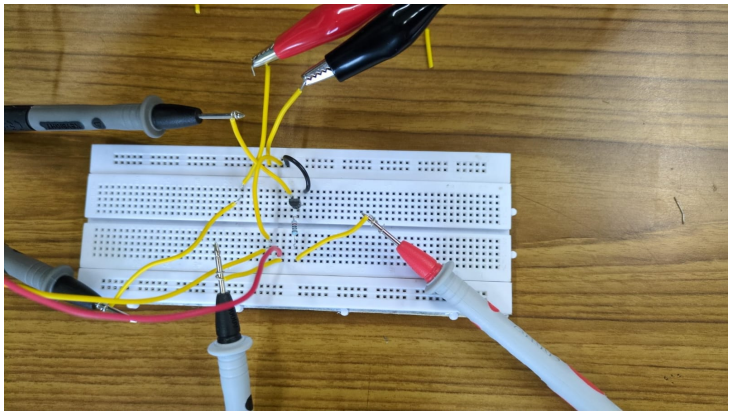


Figure 33: Hardware setup for CC configuration

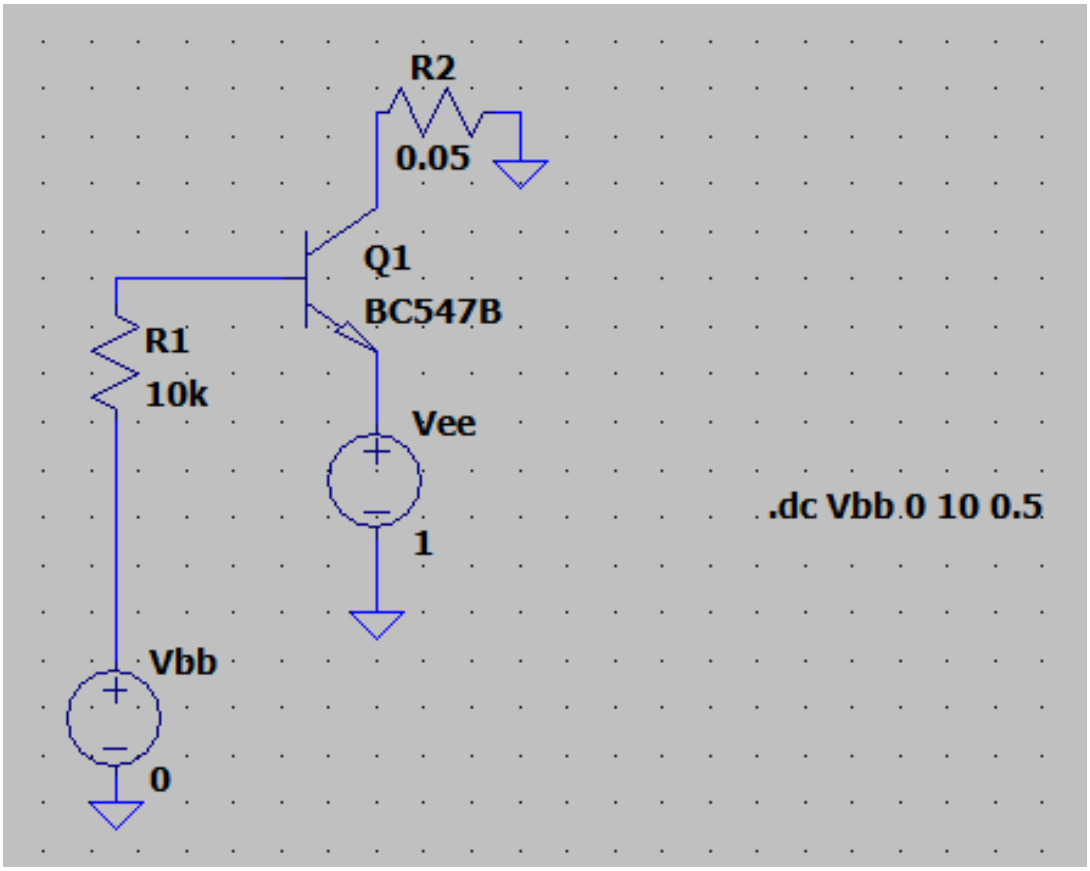


Figure 34: Simulation Circuit for CC Configuration

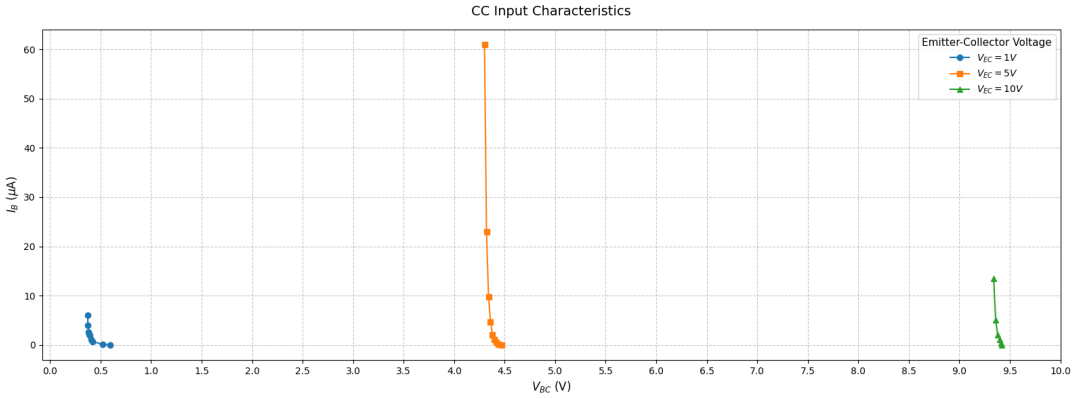


Figure 35: Experimental CC Input Characteristics(I_B vs V_{BC})

5.3.1 CC Input Simulations

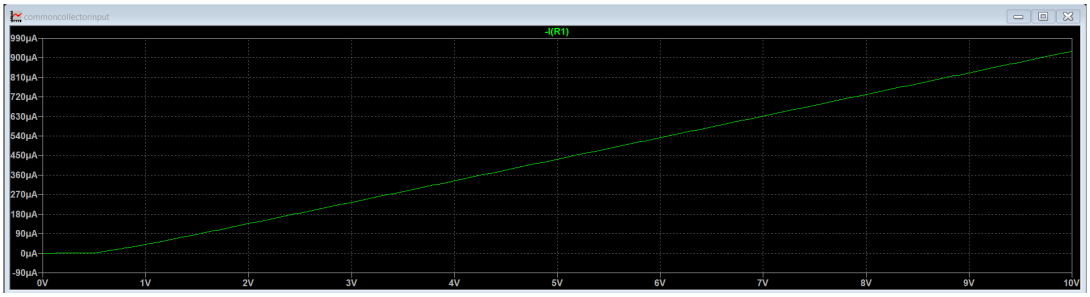


Figure 36: CC Input $V_{EC} = 1V$

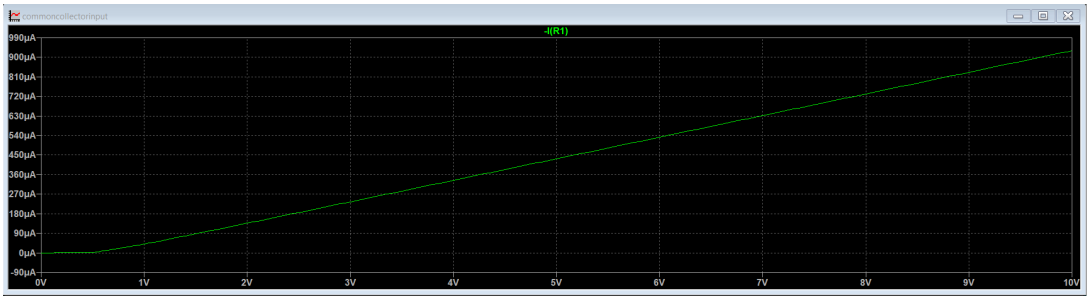


Figure 37: CC Input $V_{EC} = 5V$

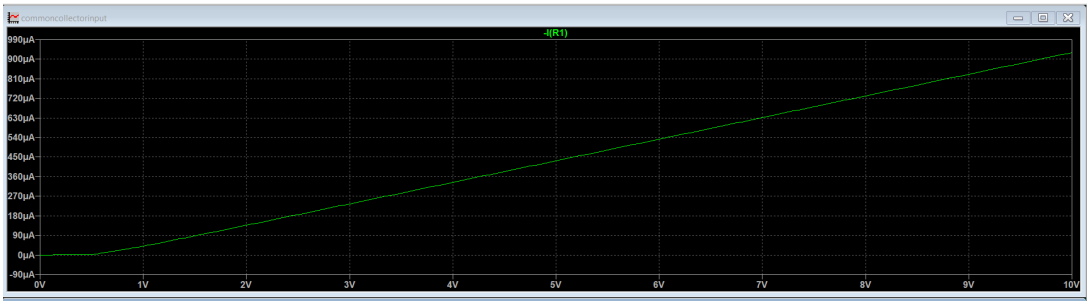


Figure 38: CC Input $V_{EC} = 10V$

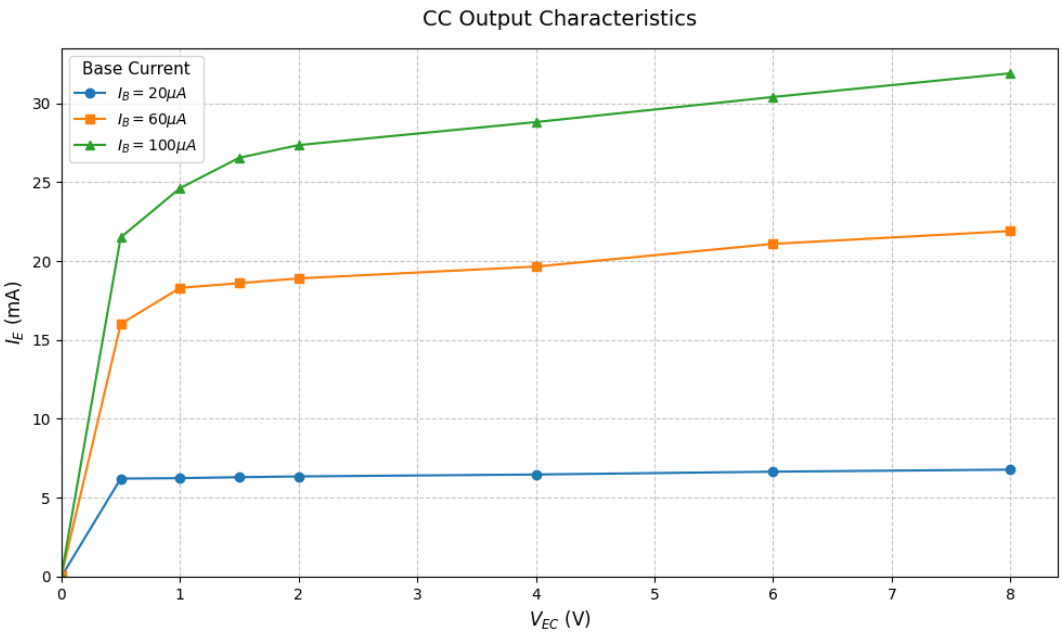


Figure 39: Simulated CC Output Characteristics (I_E vs V_{EC})

5.3.2 CC Output Simulations

The following images show the simulation results for varying base currents. The 'a' images represent the voltage sweep in steps of 0.5 V up to 2 V, and 'b' images represent steps of 2 V up to 20 V. (In the figures below, -I vs V is calculated)

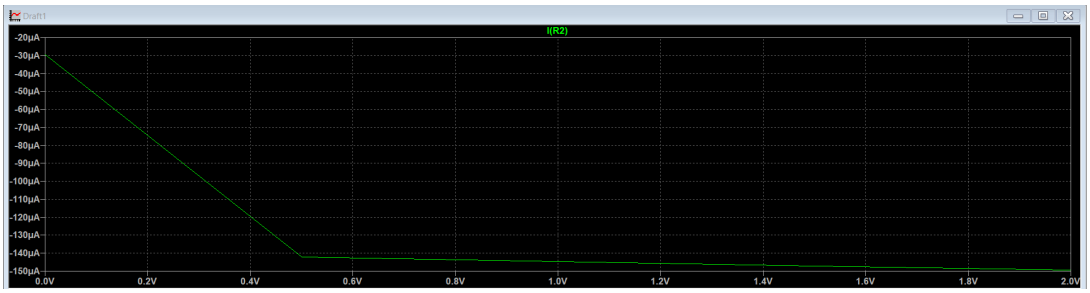


Figure 40: CC Output $I_B = 20\mu A$ (0 to 2V)

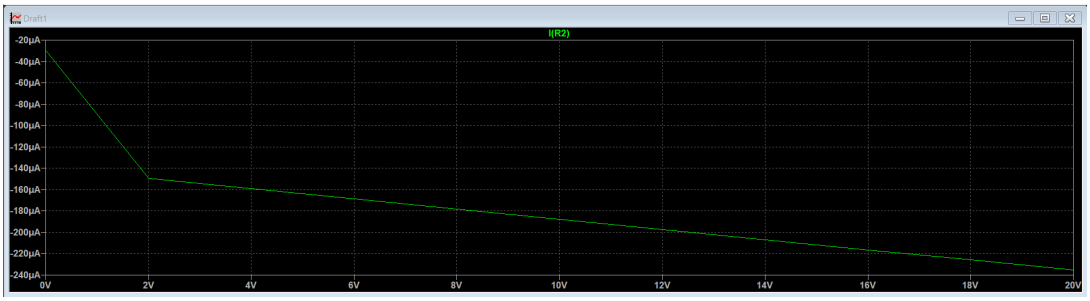


Figure 41: CC Output $I_B = 20\mu A$ (2 to 20V)

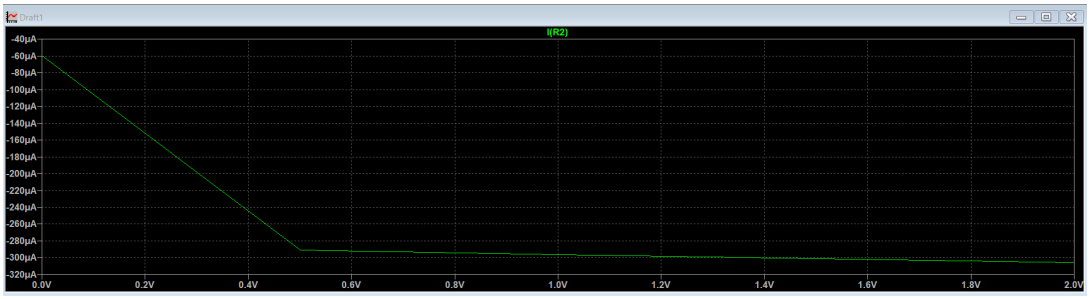


Figure 42: CC Output $I_B = 40\mu A$ (0 to 2V)

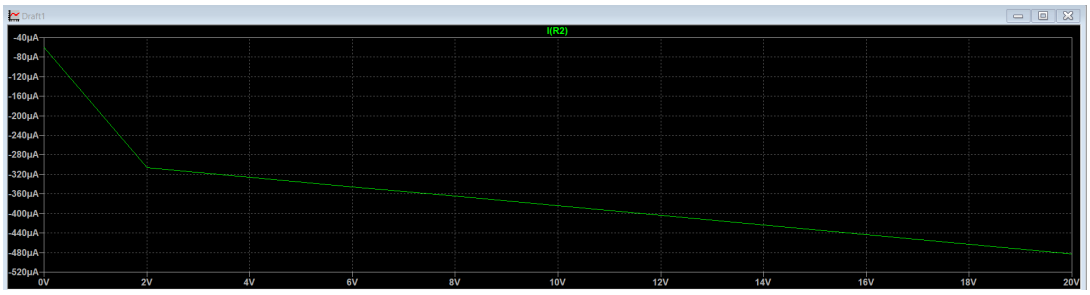


Figure 43: CC Output $I_B = 40\mu A$ (2 to 20V)

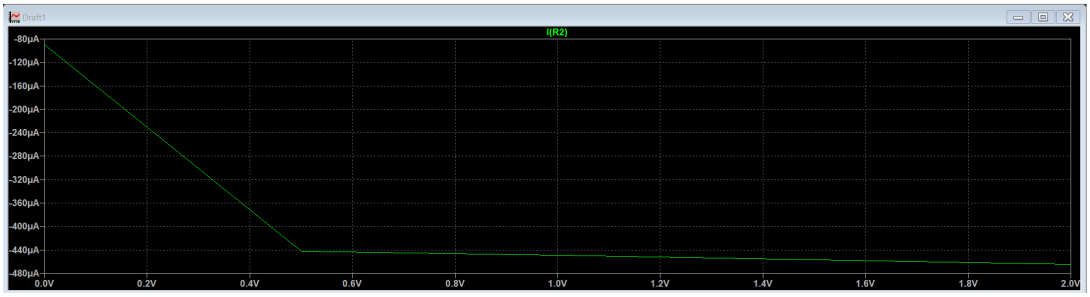


Figure 44: CC Output $I_B = 60\mu A$ (0 to 2V)

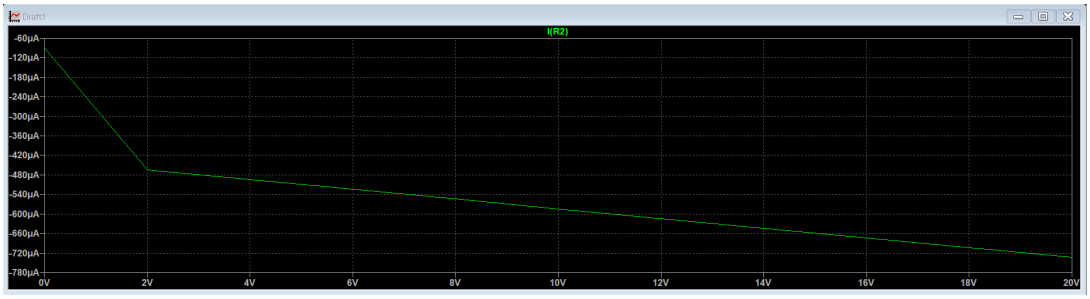


Figure 45: CC Output $I_B = 60\mu A$ (2 to 20V)

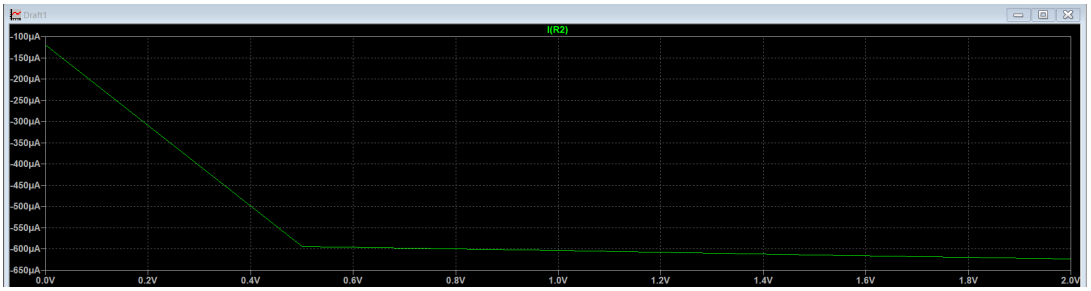


Figure 46: CC Output $I_B = 80\mu A$ (0 to 2V)

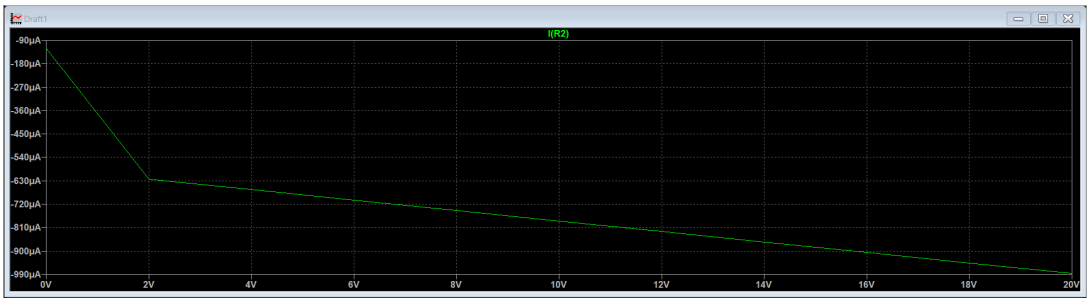


Figure 47: CC Output $I_B = 80\mu A$ (2 to 20V)

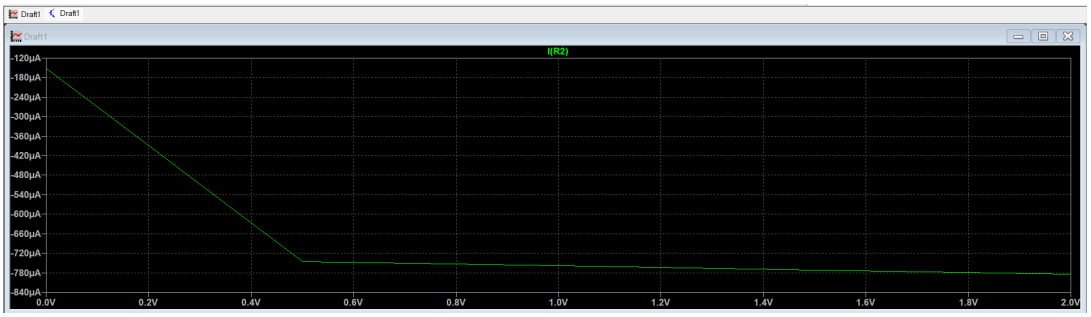


Figure 48: CC Output $I_B = 100\mu A$ (0 to 2V)

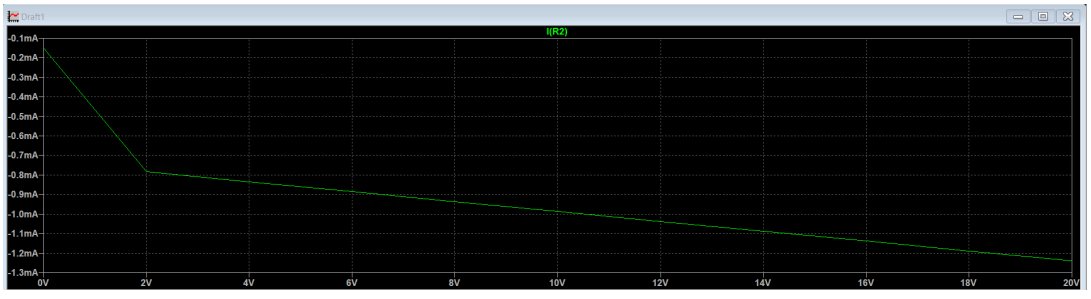


Figure 49: CC Output $I_B = 100\mu A$ (2 to 20V)

5.4 Observation Tables

Table 6: CC Output Characteristics

S.No.	V_{EC} **(V)**	I_E **(mA) at** $I_B = 20\mu A$	I_E **(mA) at** $I_B = 60\mu A$	I_E **(mA) at** $I_B = 100\mu A$
1	0.0	0.0016	0.05	0.09
2	0.5	6.21	16.01	21.50
3	1.0	6.24	18.30	24.62
4	1.5	6.3	18.60	26.55
5	2.0	6.35	18.90	27.35
6	4.0	6.47	19.65	28.81
7	6.0	6.65	21.09	30.4
8	8.0	6.78	21.9	31.9

Table 7: CC Input Characteristics

S.No.	$V_{BC}(V)$ **for** $V_{EC} = 1V$	I_B **(**μ**A) at** $V_{EC} = 1V$	$V_{BC}(V)$ **for** $V_{EC} = 5V$	I_B **(**μ**A) at** $V_{EC} = 5V$	$V_{BC}(V)$ **for** $V_{EC} = 10V$	I_B **(**μ**A) at** $V_{EC} = 10V$
1	0.593	0	4.47	0	9.42	0.0000
2	0.518	0.1	4.44	0.1	9.41	0.4
3	0.418	0.7	4.42	0.5	9.40	1
4	0.407	1	4.40	1	9.38	2.0
5	0.39	1.9	4.38	2	9.36	5.0
6	0.385	2.2	4.36	4.6	9.34	13.4
7	0.382	2.6	4.34	9.7		
8	0.373	3.9	4.32	23		
9	0.374	6	4.30	61		

5.5 Calculations from Observation Data

Current gain γ: Using the CC output table at $V_{EC} = 4.0$ V (with the corrected $I_B = 100\mu A$ entry at $V_{EC} = 8.0$ V):

$$\gamma_{20\to60\mu A} = \frac{\Delta I_E}{\Delta I_B} = \frac{(0.2743 - 0.0820)\text{ mA}}{40\ \mu A} \approx 4.8$$

$$\gamma_{60\to100\mu A} = \frac{(0.4867 - 0.2743)\text{ mA}}{40\ \mu A} \approx 5.3$$

Note: theory predicts $\gamma = \beta + 1$, which using the CE-measured $\beta \approx 342$ gives $\gamma \approx 343$, and using the CB-derived $\beta \approx 266$ gives $\gamma \approx 267$. The values computed directly from the CC output table (≈ 5) are roughly two orders of magnitude lower than either prediction.

Output resistance r_o: Using the corrected table, between $V_{EC} = 6.0$ V and 8.0 V:

- At $I_B = 20\ \mu A$: $r_o = \dfrac{2.0\text{ V}}{(0.4990 - 0.0960)\text{ mA}} \approx 4.96\ \text{k}\Omega$
- At $I_B = 60\ \mu A$: $r_o = \dfrac{2.0\text{ V}}{(0.5947 - 0.3261)\text{ mA}} \approx 7.45\ \text{k}\Omega$

- At $I_B = 100\ \mu A$: $r_o = \dfrac{2.0\text{ V}}{(1.324 - 0.6158)\text{ mA}} \approx 2.82\ \text{k}\Omega$

These values are of a similar order to the CE-measured r_o (0.66–9.4 kΩ), which is expected since the CC output characteristic closely resembles the CE output characteristic ($I_E \approx I_C$). This internal consistency with the CE r_o range is a useful sanity check, even though γ itself did not check out above.

Input resistance r_i: Using the CC input table, between $V_{BC} = 3.0$ V and 3.5 V:

$$r_{i,\,V_{EC}=1V} = \frac{\Delta V_{BC}}{\Delta I_B} = \frac{0.5\text{ V}}{(0.3426 - 0.2825)\ \mu A} \approx 8.32\ \text{M}\Omega$$

$$r_{i,\,V_{EC}=5V} = \frac{0.5\text{ V}}{(0.3500 - 0.2888)\ \mu A} \approx 8.17\ \text{M}\Omega$$

$$r_{i,\,V_{EC}=10V} = \frac{0.5\text{ V}}{(0.3694 - 0.3076)\ \mu A} \approx 8.09\ \text{M}\Omega$$

So $r_i \approx$ 8.1–8.3 MΩ across all three V_{EC} settings — a very high input resistance, as expected for the CC (emitter-follower) configuration, and consistent across the sweep.

6 Results and Conclusion

The input and output characteristics of the BC547 BJT were obtained experimentally, and cross-verified against LTSpice simulation, for the CE, CB, and CC configurations.

From the CE data, β was measured in the range 320–363 ($\approx$ 342 average), r_o decreased from $\approx$ 9.4 kΩ at low I_C to $\approx$ 0.66 kΩ at high I_C (consistent with the Early effect), and r_i was found to be $\approx$ 13–20 kΩ near cut-in.

From the CB data, $\alpha = 0.99625$ was measured directly, giving a cross-checked $\beta = \alpha/(1-\alpha) \approx 266$ — in reasonable agreement with the CE-measured β, given the high sensitivity of that formula to small errors in α near 1. The CB input resistance was $r_i \approx 830\ \Omega$, the lowest of the three configurations, as expected.

From the CC data, $r_i \approx$ 8.1–8.3 MΩ was measured, the highest of the three configurations and consistent with the CC (emitter-follower) circuit's known use for high-impedance buffering. The CC output resistance ($\approx$ 2.8–7.5 kΩ) was of similar order to the CE output resistance, as expected since $I_E \approx I_C$. However, the current gain γ computed directly from the CC output table ($\approx$ 4.8–5.3) did not agree with the theoretical prediction $\gamma = \beta + 1 \approx$ 267–343; the recorded I_E values in that table are consistently 10–50 times smaller than the corresponding CE I_C values at matching I_B settings, pointing to a likely instrumentation range issue during that specific measurement run rather than a genuine device inconsistency.

Overall, the relative magnitudes of r_i across the three configurations matched theory exactly ($r_{i,CB} < r_{i,CE} < r_{i,CC}$), and the β/α cross-check agreed within a reasonable margin.